Tvist.ai — Executive Summary

The settlement-trust layer for agentic commerce: jurisdiction, consent, escrow, and recall for AI agents that move money.

THE PROBLEM

AI agents now spend money on people's behalf — ~1 in 6 Black Friday 2025 purchases were AI-assisted; gen-AI retail traffic +4,700% YoY (Adobe); Juniper names **trust the #1 barrier** to agentic commerce. **And the rails they spend on are instant and irrevocable by regulation** — Pix (MED 2.0), SEPA Instant (10-second finality), FedNow, UPI; the ECB warns instant-payment fraud risk is "up to 10× higher." Dispute economics were already broken on cards: 106M Visa disputes in 2025 (→ ~324M by 2028), 40–80% friendly fraud, ~$28B/yr merchant losses. **When an agent pays the wrong party, exceeds its mandate, or prepays for a service that never arrives — on a settlement-final rail — there is no recourse layer. Nobody owns the cross-section.**

OPTIONS IN THE MARKET — AND THEIR GAPS

Player	Solves	Gap
Visa AI dispute suite (6 tools)	Card-dispute automation at network scale	Card-only, single network
Mastercard (Mastercom, Ethoca, VI, Agent Pay)	Dispute APIs + open intent standard	Protocols, not operations — no cross-rail orchestration, no escrow
Amex Agent Purchase Protection	Liability cover for agent purchases	Closed-loop, Amex-only
Agent-payment protocols (AP2, ACP, OKX APP, x402)	Agent checkout & payment execution	Disputes "future scope" or absent
Chargeback SaaS (Chargeflow, …)	Card representment for merchants	Nothing for irrevocable A2A or agent mandates
Escrow / custody (Modulr, Trustly, …)	Licensed fund-holding per rail	No programmable, dispute-aware, rail-agnostic escrow

Seven agent-payment protocols, four networks, ~$50B of manual escrow — everyone built rails and intent standards; nobody built the dispute + escrow layer across them.

THE TVIST.AI SOLUTION — FOUR PRIMITIVES AROUND EVERY AGENT TRANSACTION

1 · Jurisdiction, agreed up front. From a global registry of 22 real instant/A2A regimes, both parties negotiate the governing dispute regime before funds move — selected as the **Nash bargaining optimum**. No overlap → no transaction. **2 · Consent, enforced.** The principal's mandate (budget, allowlist) is checked *before* settlement, not refunded after. **3 · Escrow, delivery-gated.** Funds hold until a typed condition (delivery / attestation / time) is provably met. **4 · Recall, proof-gated.** A settled payment reverses only where the agreed regime permits, in-window, on evidence of a mandate breach — unilateral clawbacks refused, exactly as irrevocable-rail rulebooks require. **+ Grounded in law:** every code maps to one of 7 civil/commercial-law categories; all 22 jurisdictions carry their operative statutes, scheme rulebooks & regulator linked to the official source — an accepted dispute returns the **citation inline** (BGB §177 falsus procurator in Brazil; UCC §2-711 in India). **+ x402 on-rail:** native HTTP-402 agent payments (signed X-PAYMENT → resource + receipt), replay-safe, consent-capped even on-chain. Tvist is a thin orchestration layer — not a bank; regulated custody partners hold funds.

ARCHITECTURE — SUPER HIGH LEVEL

HUMANS — browser, console	AI AGENTS — curl / SDK / SKILL.md
ONE DUAL-USE API · content-negotiated · self-describing	
1 · JURISDICTION 22-regime registry · Nash-optimal selection	
2 · TRUST consent / intent vault · mandate checks	
3 · VALUE conserved ledger · escrow hold / release	
4 · RESOLUTION regime-gated recall · law-cited disputes	
custody partners (funds)	rails: Pix · SEPA · FedNow · UPI · FPS · NPP · M-Pesa · x402 stablecoin …

PROOF, TODAY (NOT SLIDEWARE)

Adversarially validated in MIT/NANDA's Nanda Town rig: 4 swarm scenarios, 10 adversarial validators — every attack (unilateral clawback, escrow drain, over-mandate spend, friendly-fraud refund, off-regime dispute) passes against a plain ledger and is **blocked by Tvist** on identical scenarios; 535 tests green. **Live dual-use service**: hosted API + animated site with in-page playground; 25 endpoints incl. legal taxonomy + x402; one SKILL.md; 49 endpoint tests, 584 green across the project. **Flagship use case**: DigiDoot — "a personal AI agent for every Indian citizen" — maps 1:1 onto Tvist as its settlement endpoint (consent → intent vault, UPI → regime, journeys → escrow/recall).

POSITIONING

Complementary, not competitive: to **networks** (we operationalize their intent standards), to **custody providers** (we are their missing rules engine), to **agent platforms** (we are the trust layer their checkout protocols defer). The wedge is the cross-section nobody owns — and it compounds with every rail, protocol, and jurisdiction added to the registry.

Disclaimer: technical demonstration — a notional-credits sandbox, not a bank, payment institution, or law firm. Legal references cite real primary sources but are not legal advice; obtain qualified local counsel before relying on them in any jurisdiction.